

Entropy Across Singular Projections: Fiber Multiplicity, Caustics, and Observer-Relative Information

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A lower-dimensional observer who watches a hidden dynamical system through a projection must count, distinguish, and assign probability to the hidden states compatible with each observation. When the map from the hidden evolution parameter to observed time has singularities, that counting changes abruptly. This paper treats the simplest such singularity, the fold, at finite observational resolution. We prove a local theorem stating that a nondegenerate fold changes the number of hidden preimages by two while an orientation-weighted count is conserved. We compute closed-form branch entropies for a family of canonical control maps. The quadratic fold carries exactly one bit at every two-branch slice, the symmetric slice of a double fold carries exactly one and one half bits, and a degenerate critical point of a monotone map carries no entropy at all, so multiplicity rather than criticality is what stores the information. We compute the resolution scaling of the binned entropy of the fold caustic and verify numerically that it converges to its differential limit with deviations shrinking as the square root of the bin width. Finally we demonstrate, in a reversible Hamiltonian model, that the observed binned entropy of the coordinate-time marginal changes by more than two bits along a trajectory while momentum reversal recovers the initial hidden ensemble to near machine precision, so the observed entropy change is produced and destroyed by folds of the projection with zero hidden information loss. Nothing in this paper is a claim about physical spacetime, about the thermodynamics of real systems, or about quantum gravity.

I. INTRODUCTION

An observer rarely sees the full state of the system it observes. It sees a projection, and it parameterizes what it sees by its own time. If the hidden evolution parameter and the observed time increase together everywhere, the observer's bookkeeping is simple. Each observed instant corresponds to one hidden state, and observed uncertainty tracks hidden uncertainty. The interesting case is the other one. If the observed time, viewed as a function of the hidden evolution parameter, has a critical point, then a single observed instant can correspond to several hidden states at once. The set of hidden states compatible with one observation, which we call the fiber of that observation, changes its size discontinuously as the observed time crosses the critical value.

This paper studies how a finite-resolution observer counts and weighs the members of such fibers, and how much information the observer's ignorance is worth, in the simplest singular situation. That situation is the fold, the generic critical point of a map from one dimension to one dimension. Singularity theory guarantees that folds are stable under small perturbations, so they are the correct first object of study, and it classifies everything more complicated as a higher catastrophe.

Four results are reported, and each carries an explicit evidence label. First, a local theorem, proved, describing what a fold does to the fiber. The number of hidden preimages of an observed instant jumps by two across a fold, while a signed count that weights each preimage by its time orientation does not change at all. Second,

closed-form entropy values, computed and verified in two independent implementations, for a family of canonical control maps. A quadratic fold is worth exactly one bit of branch ambiguity, a symmetric double fold is worth exactly one and one half bits, and a monotone map with a degenerate critical point is worth nothing. Third, a resolution-scaling law, computed and numerically demonstrated, for the observed density near a fold. That density diverges at the fold, but the divergence is integrable, and the binned observed entropy converges to a finite differential limit as the bin width shrinks, with deviations falling as the square root of the bin width. Fourth, a numerical demonstration in a small Hamiltonian model. A reversible hidden dynamics, integrated with a time-symmetric method, produces observed entropy changes of more than two bits along a trajectory, and then gives every bit back. Reversing the momenta recovers the initial hidden ensemble to near machine precision, and the entropy curve retraces itself exactly. The observed entropy change is therefore a property of the projection and the observer's resolution, not of any loss in the hidden dynamics.

The scope of these claims is deliberately narrow. The theorem is a statement in one-dimensional singularity theory. The closed-form values are exact computations for named control maps. The demonstration is a simulation of a toy model. No claim is made about physical spacetime, about the thermodynamics of any real system, or about quantum gravity. The paper does establish, within its model class, a clean separation between two things that are often conflated, the growth of an observer's coarse-grained entropy and the destruction of hidden information. Folds of a projection produce the first without any of the second.

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II. SETUP

Let $\mathcal{M}$ be a smooth manifold, the hidden state space, and let

$$z : I \rightarrow \mathcal{M} \quad (1)$$

be a smooth trajectory parameterized by a hidden evolution variable τ on an interval I . A smooth observation map

$$\pi : \mathcal{M} \rightarrow \mathcal{X} \quad (2)$$

sends hidden states to an observed space $\mathcal{X}$, which we take to carry spacetime coordinates x^μ . An observer supplies a time functional

$$T_u(x) = u_\mu x^\mu, \quad (3)$$

with u the observer's time direction, and reads the trajectory through the scalar time map

$$f(\tau) = T_u(\pi(z(\tau))). \quad (4)$$

Definition 1 (Fiber of an observed time). *The fiber of the observed time t is the preimage*

$$F(t) = \{ \tau \in I : f(\tau) = t \}, \quad (5)$$

the set of hidden parameter values compatible with the observation t .

The unsigned multiplicity $N(t) = |F(t)|$ counts the fiber. The signed count weights each element by its time orientation,

$$N_{\text{sgn}}(t) = \sum_{\tau_i \in F(t)} \text{sgn } f'(\tau_i), \quad (6)$$

defined on regular slices, meaning values of t whose fiber contains no critical point of f .

The observer does not resolve t exactly. It resolves t to within a bin width $\epsilon > 0$, and its observed distribution is the pushforward of the hidden distribution under f , accumulated into bins of width ϵ . We write H_ϵ for the Shannon entropy in bits of that binned distribution. Finite resolution is mandatory rather than a modeling convenience. Differential entropy is not invariant under reparameterization of the observed variable, and near a singularity of f the pushforward density diverges, so an unbinned continuous entropy assigned to the observed variable depends on coordinate choices in exactly the region this paper is about. The binned quantity H_ϵ is well defined for every $\epsilon > 0$, and every entropy statement below is either a statement about H_ϵ at stated ϵ , a statement about a discrete branch-label distribution, or an exact limit as $\epsilon \rightarrow 0$ with the divergent resolution term removed.

III. THE FOLD THEOREM

The fold is the nondegenerate critical point of the time map. The statement below is kinematic. It concerns the map f alone and involves no dynamics, no probability, and no physics.

Theorem 1 (Local fold multiplicity). *Let f be smooth near τ_0 with $f'(\tau_0) = 0$ and $f''(\tau_0) \neq 0$, and let $t_0 = f(\tau_0)$. Then in suitable smooth local coordinates*

$$f(\tau) - t_0 = \sigma (\tau - \tau_0)^2, \quad \sigma \in \{+1, -1\}. \quad (7)$$

For $\sigma = +1$ the unsigned multiplicity of nearby regular slices is 0 for $t < t_0$ and 2 for $t > t_0$, and the two branches carry opposite signs of f' . The signed count N_{sgn} takes the same value on both sides of the fold. For $\sigma = -1$ the time-reversed statement holds.

Proof. The one-dimensional Morse lemma gives the normal form (7). For $\sigma = +1$ and $t > t_0$ the local preimages are $\tau_\pm = \tau_0 \pm \sqrt{t - t_0}$, with $f'(\tau_\pm) = \pm 2\sqrt{t - t_0}$, so the two branches have opposite orientation and their contributions to (6) cancel. For $t < t_0$ there are no local preimages, contributing zero. The local signed count is therefore 0 on both sides. The case $\sigma = -1$ follows by reversing the sign of the time coordinate. $\square$

The fold is stable. Whitney's classification of stable maps of the plane and the catastrophe-theory literature descending from it establish that a nondegenerate fold persists under small smooth perturbations and that its square-root branch geometry is universal [1, 2]. Those structural statements are standard mathematics and are cited, not reproved. What the present campaign adds is a numerical verification of the same statements by the instrument stack used throughout this paper, and that verification is labeled demonstrated, not proved. In five hundred perturbation trials per amplitude, the fold persisted with probability 1.000 at every perturbation norm up to 0.12. The branch-separation exponent, fitted over six decades of distance to the fold, was 0.500000000000 against the square-root law, with fit error below 10^{-10} . The deviations of the perturbed fold location and fold curvature from first-order perturbation theory scaled as the square of the perturbation amplitude across seven decades of measured deviation. At the largest tested amplitude, norm 0.32, persistence dropped to 0.956, which is the expected exit from the perturbative regime rather than an instrument failure. These numbers validate the instruments against known mathematics. They are not new mathematics.

IV. BRANCH WEIGHTS AND BRANCH ENTROPY

Suppose the hidden parameter τ is distributed uniformly on I . The coarea formula for the pushforward

of a density under a map with nonvanishing derivative on the fiber assigns each preimage $\tau_i \in F(t)$ a weight proportional to the reciprocal of the local time speed [6],

$$w_i(t) = \frac{1/|f'(\tau_i)|}{\sum_j 1/|f'(\tau_j)|}. \quad (8)$$

An observer who resolves the observed time but not the branch label is ignorant of which preimage the system occupies, and the value of that missing label is the branch entropy

$$H_{\text{br}}(t) = - \sum_i w_i(t) \log_2 w_i(t) \quad (9)$$

in bits [4, 5].

The following values are proved by direct computation from (8) and (9) and were verified in two independent implementations, one in MATLAB and one in Python, which agree on branch locations to 4.4×10^{-16} .

Quadratic fold. For $f(\tau) = \tau^2$ the two branches at any regular slice $t > 0$ sit at $\tau = \pm\sqrt{t}$ with equal speeds $|f'| = 2\sqrt{t}$. The weights are $(\frac{1}{2}, \frac{1}{2})$ and

$$H_{\text{br}}(t) = 1 \text{ bit} \quad \text{for every } t > 0. \quad (10)$$

The fold hands the observer exactly one new binary distinction, uniformly across its two-branch side.

Double fold. For $f(\tau) = \tau^3 - \tau$, at the symmetric slice $t = 0$ the fiber is $\{-1, 0, 1\}$ with speeds $|f'| = (2, 1, 2)$. The weights are $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ and

$$H_{\text{br}}(0) = \frac{3}{2} \text{ bits} \quad (11)$$

exactly. As the slice approaches a band edge, the two merging branches dominate the fiber measure because their speeds vanish, and the branch entropy tends to 1 bit. The measured deviation from 1 bit is below 10^{-3} at a distance of 10^{-9} from the edge.

Monotone maps. For $f(\tau) = \tau$, and equally for the degenerate cubic $f(\tau) = \tau^3$, every fiber is a single point and

$$H_{\text{br}}(t) = 0 \text{ bits} \quad \text{for every regular } t. \quad (12)$$

The cubic has a critical point at the origin, yet contributes no branch entropy anywhere, because the map remains monotone and no slice ever meets more than one preimage. Multiplicity, not criticality, is what carries the information. Throughout all four control maps the signed count (6) is constant on regular slices, so the observational multiplicity and its entropy grow while the topological invariant is conserved, exactly as Theorem 1 requires.

V. THE FOLD CAUSTIC AND RESOLUTION SCALING

Where the derivative of the time map vanishes, the projection compresses hidden measure into a small observed interval, and the observed density diverges. For

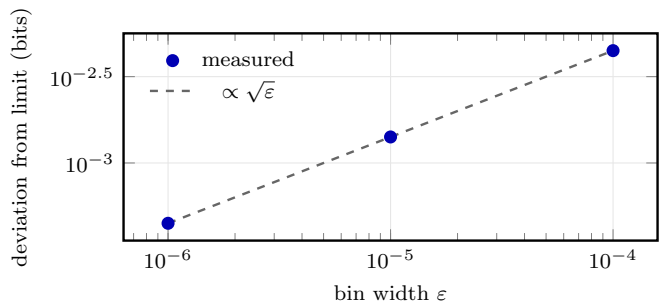


FIG. 1. Convergence of the binned fold-caustic entropy to its differential limit. The deviation of $H_\epsilon + \log_2 \epsilon$ from $1 - 1/\ln 2$ bits shrinks proportionally to the square root of the bin width across the tested range, so the inverse-square-root caustic is integrable and produces no entropy pathology. Points are recomputed deterministically from the sealed instrument.

the quadratic fold $f(\tau) = \tau^2$ with τ uniform on $[-1, 1]$, the pushforward density is

$$p(t) = \frac{1}{2\sqrt{t}}, \quad 0 < t \leq 1, \quad (13)$$

the inverse-square-root divergence familiar from caustics in optics, where a family of rays folds onto an envelope and the intensity diverges with the same exponent at the same codimension [3]. The divergence in (13) is integrable, so the fold concentrates observed probability without creating any entropy pathology.

The binned observed entropy of (13) obeys a clean resolution law. A direct computation of the differential entropy of (13) gives $1 - 1/\ln 2$ bits, approximately -0.4427 bits, the negative value being ordinary for a density concentrated on a subunit interval. The binned entropy at width ϵ then satisfies

$$H_\epsilon + \log_2 \epsilon \rightarrow 1 - \frac{1}{\ln 2} \quad (\epsilon \rightarrow 0), \quad (14)$$

with the bin masses computed exactly from the cumulative distribution $\sqrt{t}$. The measured deviations from the limit are 0.00447 at $\epsilon = 10^{-4}$, 0.00141 at $\epsilon = 10^{-5}$, and 0.000447 at $\epsilon = 10^{-6}$, shrinking as $\sqrt{\epsilon}$, the rate set by the single bin that straddles the divergence. The convergence (14) is the quantitative sense in which the caustic is harmless. All resolution dependence sits in the universal term $-\log_2 \epsilon$ shared by every absolutely continuous observed density, and the fold contributes only a finite, computable offset. This convergence was verified in the stated range and is labeled demonstrated there, with the closed-form limit computed exactly.

VI. A REVERSIBLE FOLD CYCLE

The results above are static. This section demonstrates, in a model, what folds do to an observer's entropy along

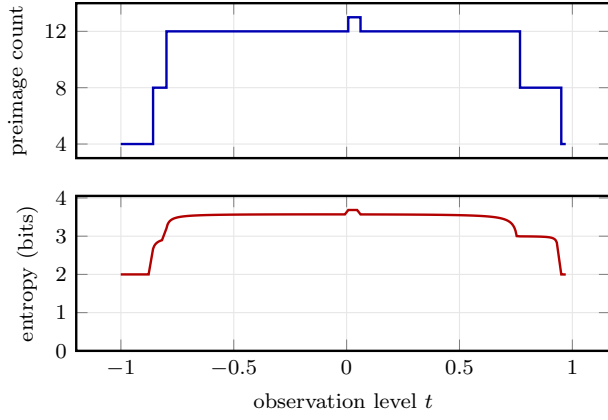


FIG. 2. The observation axis of one hidden trajectory of the toy Hamiltonian. The unsigned preimage count (top) is a staircase whose breakpoints are the fold values of the observed time and whose interior band profile is 4, 8, 12, 13 and then the same values descending, constant between breakpoints, and the signed count (6) obeys the path degree rule at each of the roughly 370 tested levels. That rule states that the signed crossing count of a level equals the sign of the net displacement of t when the level lies between the endpoint values of t and equals zero otherwise. One caveat is recorded with this result rather than elsewhere. The twelve fold values of t cluster near the turning amplitudes of the trajectory, so only the two isolated breakpoints, the endpoint values with steps of one, admitted the per-breakpoint step check at this level resolution, while the clustered fold breakpoints were verified in aggregate through steps of four across close pairs together with band constancy. A finer level grid around the clusters is the designed follow-up.

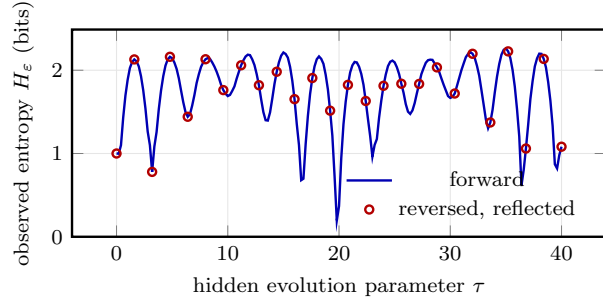


FIG. 3. Observed binned entropy of the coordinate-time marginal for the ten-thousand-member ensemble. The forward curve spans 2.07 bits. The open circles show the momentum-reversed run reflected in time, which retraces the forward curve with defect exactly zero while the initial hidden ensemble is recovered to 1.1×10^{-14} , so the entire entropy change is observational.

a dynamical trajectory, and what they do not do. Everything in this section carries the label demonstrated in a model, at exploratory grade, and nothing in it is a thermodynamic claim.

The hidden dynamics is the conserved toy Hamiltonian

$$H = \frac{1}{2}(p_t^2 + p_x^2 + p_u^2) + \frac{1}{2}\omega_t^2 t^2 + \frac{1}{2}\omega_u^2 u^2 + \frac{\lambda}{4}u^4 + gEtu, \quad (15)$$

with $\omega_t = 1$, $\omega_u = 1.2$, $\lambda = 0.1$, $g = 0.25$, and $E = 1$. The coordinate t plays the role of observed time, with $dt/d\tau = p_t$, the coordinate u is hidden, and folds of the time map occur wherever p_t crosses zero with nonzero acceleration. The integrator is velocity Verlet with step 10^{-3} over $\tau \in [0, 40]$, a symplectic and time-symmetric method, and the relative energy drift over the full run was 6.1×10^{-9} , within the sealed fixed-step bar of the

tolerance freeze described in Section VIII. The model (15) is not relativistic and is not proposed as a theory of anything. It is an instrumented example whose only job is to produce genuine folds under a genuinely reversible dynamics.

Single-trajectory audit. One trajectory of (15) produces twelve folds of the time map. Observed through constant-time levels, its unsigned multiplicity is a staircase over the observation axis with band profile 4, 8, 12, 13 and then the same values descending, constant between breakpoints, and the signed count (6) obeys the path degree rule at each of the roughly 370 tested levels. That rule states that the signed crossing count of a level equals the sign of the net displacement of t when the level lies between the endpoint values of t and equals zero otherwise. One caveat is recorded with this result rather than elsewhere. The twelve fold values of t cluster near the turning amplitudes of the trajectory, so only the two isolated breakpoints, the endpoint values with steps of one, admitted the per-breakpoint step check at this level resolution, while the clustered fold breakpoints were verified in aggregate through steps of four across close pairs together with band constancy. A finer level grid around the clusters is the designed follow-up.

Reversible ensemble. An ensemble of 10^4 members, initialized from a fixed seeded Gaussian spread and integrated with the same method, gives the observed side of the demonstration. The observed quantity is the binned entropy H_e of the coordinate-time marginal, with bin width 0.05 on the range $[-3, 3]$. Along the forward trajectory that entropy spans 2.0708 bits, from 0.1952 bits to 2.2660 bits, rising and falling as fold bands raise and lower the fiber multiplicity. The hidden side of the demonstration is the reversal. Flipping every momentum at the final time and integrating forward again recovers the initial ensemble with maximum residual 1.1×10^{-14} over all members and coordinates, and the reversed entropy curve retraces the forward curve with defect exactly zero, a bit-identical retrace guaranteed by the time symmetry of the integrator. The falsification bar for this run, sealed in advance, would have been tripped by any signed-count violation, any failure of band constancy, a recovery residual above 10^{-9} , or a reversed entropy curve that failed to retrace the forward one. None tripped.

The conclusion is stated exactly. In this model, more than two bits of observed entropy appear and disappear along a trajectory whose hidden dynamics loses nothing. The observed entropy change is observational. It is produced and destroyed by folds of the projection together with the observer's finite resolution, with zero hidden information loss.

Three distinct quantities must not be conflated in reading that sentence. The first is the conditional entropy of the hidden state given the observation, the observer's residual uncertainty, which folds change by changing the number and weighting of compatible hidden states. The second is the coarse-grained entropy of the observed, binned distribution, the quantity measured above, which

TABLE I. Entropy signatures by singularity class. Rows above the rule are established in this paper at the stated labels. Rows below the rule are open questions stated as design targets.

Class	Multiplicity	Signed count	Entropy signature
Regular	constant	conserved	none
Degenerate	constant	conserved	none
monotone			
Fold	jumps by two	conserved	one new binary distinction
Cusp	varies across fold boundaries	conserved globally	correlated branch ambiguity (open)
Higher catastrophes	larger structured changes	constrained	multiway ambiguity (open)

a reversible hidden dynamics can raise and lower freely. The third is thermodynamic entropy production, which requires a physical mechanism that makes lost distinctions inaccessible, such as dissipation, an unmonitored environment, or the erasure of a memory, and which carries the Landauer cost [7, 8]. Nothing in this paper measures or bounds the third quantity. The demonstration shows precisely that the first two can undergo large changes while the third has no mechanism to occur, since the dynamics is reversible and reversed.

VII. ENTROPY SIGNATURES OF PROJECTION SINGULARITIES

The results above organize into a classification of what each local type of projection singularity contributes to an observer’s information budget. Table I states it. The fold row and the two regular rows are established by the preceding sections at the labels given there. The cusp and higher rows are design, not results. They state the

predictions this framework makes and the experiments that would test them, and no measurement reported here bears on them.

The organizing statement is the pairing of the fold row with the degenerate-monotone row. A fold gives the observer exactly one new binary distinction, worth one bit at equal weighting and never more than one bit, while a degenerate critical point of a monotone map gives the observer nothing at all. Criticality of the time map is therefore neither necessary nor sufficient for observational entropy. What is both necessary and sufficient, in the one-dimensional setting treated here, is a change in fiber multiplicity, and Theorem 1 fixes the quantum of that change at two while conserving the signed count. Whether the cusp contributes a correlated two-bit structure, and whether the higher catastrophes contribute the multiway ambiguities their normal forms suggest, are the designed continuations, and they are listed in the table as open.

VIII. METHODS AND REPRODUCIBILITY

All instruments used in this paper were validated against exact analytic controls before any result-bearing run, under a sealed tolerance freeze registered as PF0-FREEZE-001, which fixes every numerical bar in advance and voids itself if edited. The control net comprises a monotone null, the exact quadratic creation and annihilation folds, the degenerate cubic, and the double fold, in both a MATLAB implementation and an independent Python implementation. Cross-language branch locations agree to 4.4×10^{-16} , and the same scenario executed on two substrates yields bit-identical scenario and outcome hashes. Every result-bearing run writes an append-only JSON evidence record in the repository containing the complete parameter set, seeds, outcome arrays, and SHA-256 hashes of inputs and outputs, and the records cited here carry the exploratory label of the campaign’s evidence-grading scheme. A reducer may summarize records but may not rewrite them.

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